FORENSICS MEDIUM

Fallen Leaf

Extracted the raw ext4 image `leafdisk.img` from `leafdisk.img.xz`.

Instead of mounting it (which can mess with timestamps/inodes), poked inside directly with `debugfs`:

```
$ debugfs -R 'cat /home/leaf/.bash_history' leafdisk.img for f in k_*.txt; do gzip -N $f; done openssl enc -aes-256-cbc -pbkdf2 -iter 100000 -salt -in harvest.txt -out Documents/harvest.enc -pass file:.cache/k_active shred -u harvest.txt rm -f .cache/k_* history -c
```

The user shredded `harvest.txt` and deleted the key files. But they left a note in `Documents/notes.txt`:

```
$ debugfs -R 'cat /home/leaf/Documents/notes.txt' leafdisk.img Rotating keys again. Active key is the one whose sha256 is d389cee5d239c721a5c0967d956d0826d182ca3f537a6f45b5ba40386c0a4c55
```

Ran `lsdel` in debugfs to see deleted inodes:

```
$ debugfs -R 'lsdel' leafdisk.img
```

Found 4 deleted inodes: 24, 25, 26, 27

Dumped and un-gzipped all four:

```
$ for i in 24 25 26 27; do debugfs -R "dump <$i> $i.gz" leafdisk.img gzip -dc "$i.gz" done leafkey:Gr33n-Sapl1ng-77 leafkey:R00t-B4ll-Aut0mn-12 leafkey:Phl03m-X7lem-Fl0w-90 leafkey:C4mb1um-R1ng-2024
```

Hashed them — `Phl03m-X7lem-Fl0w-90` matched the SHA256 in `notes.txt`.

Dumped `harvest.enc` and decrypted it with OpenSSL:

```
$ debugfs -R 'dump /home/leaf/Documents/harvest.enc harvest.enc' leafdisk.img $ openssl enc -d -aes-256-cbc -pbkdf2 -iter 100000 -in harvest.enc -pass pass:Phl03m-X7lem-Fl0w-90
```

Decrypted content:

Solution Script: `Fallen_Leaf.py` (included in attached solution archive)

First flag was a decoy. Real flag is split into two:

- Part 1: `d3l3t3d_k3ys_n3v3r_d13`
- Part 2: stored in the `leaf` extended attribute on `notes.txt`.

Dumped the xattr:

```
$ debugfs -R 'ea_list /home/leaf/Documents/notes.txt' leafdisk.img user.leaf = "L52GQM27PBQXI5DSL5ZDG3JTNVRDG4TT"
```

```
$ echo -n "L52GQM27PBQXI5DSL5ZDG3JTNVRDG4TT" | base32 -d _th3_xattr_r3m3mb3rs
```

```
[+] Carved notes.txt:
Rotating keys again. Active key sha256: d389cee5d239c721a5c0967d956d0826d182ca3f537a6f45b5ba40386c0a4c55
Wrote it in gz form to rotate; cleaned up after.

[+] Found user.leaf extended attribute (Base32): L52GQM27PBQXI5DSL5ZDG3JTNVRDG4TT
```

Triage and Attribute Carving

Running `solve.py` to combine both halves:

```
$ python3 solve.py
```

```
~/SecLeaf writeups/repo_clone/SecLeafCTF/FORENSICS/Fallen_Leaf main
❯ python3 solve.py
Part 1: d3l3t3d_k3ys_n3v3r_d13
Part 2: _th3_xattr_r3m3mb3rs
Flag: SecLeaf{d3l3t3d_k3ys_n3v3r_d13_th3_xattr_r3m3mb3rs}
```

Flag Proof

Combined both halves:

Solution Script: Fallen_Leaf.py (included in attached solution archive)

RECOVERED FLAG **SecLeaf{d3l3t3d_k3ys_n3v3r_d13_th3_xattr_r3m3mb3rs}**